OPERATIONS MANAGEMENT

UNIT – V

RESOURCE REQUIREMENT PLANNING

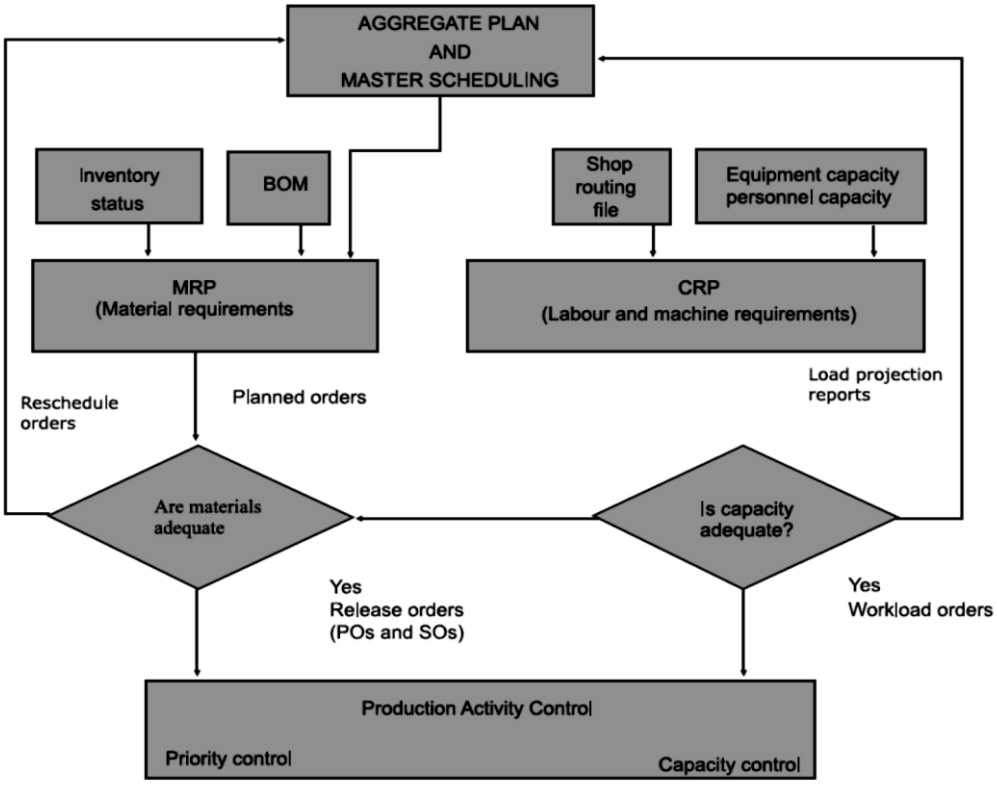
Material Requirements Planning (MRP) is a computer-based technique for determining the quantity and timing for the acquisition of dependent demand items needed to satisfy the master schedule requirements. By identifying precisely what, how many, and when components are needed, MRP systems are able to reduce inventory costs improve scheduling effectiveness, and respond quickly to market changes.

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Capacity Requirements Planning (CRP) is the process of determining what personnel and equipment capacities (times) are needed to meet the production objectives embodied in the master schedule and the material requirements plan.

The MRP tells companies when to order the materials so they can make the products to fulfil demand. CRP tells companies how many products they can make per employee, per workstation, etc. per hour, per day, per month.

"MRP focuses upon the priorities of materials, whereas CRP focuses primarily upon time".



Material and Capacity Planning Flow chart

Material Requirements Planning Objectives

1. **Inventory reduction:** MRP determines how many components are required when they are required in order to meet the master schedule. It helps to procure the materials/ components as and when needed and thus avoid excessive build-up of inventory.

2. **Reduction in the manufacturing and delivery lead times:** MRP identifies materials and component quantities, timings when they are needed, availabilities and procurements and actions required to meet delivery deadlines. MRP helps to avoid delays in production and priorities production activities by putting due dates on customer job order.

3. **Realistic delivery commitments:** By using MRP, production can give marketing timely information about likely delivery times to prospective customers.

4. **Increased efficiency:** MRP provides a close coordination among various work centres and hence help to achieve uninterrupted flow of materials through the production line. This increases the efficiency of production system.

MRP INPUTS AND OUTPUTS:

The essential inputs and outputs in an MRP system are listed below:

Inputs

• MPS of end items required.

• Inventory status file of on-hand and on-order items, lot sizes, lead times, etc.

• Product structure (BOM) file of what components and subassemblies go into each end product.

Outputs

• Order release data to CRP for load profiles

• Orders to purchasing and in-house production shops.

• Rescheduling data to MPS.

• Management reports and inventory updates

The inputs to the MRP system are: (1) A master production schedule, (2) An inventory status file and (3) Bill of materials (BOM).

MASTER PRODUCTION SCHEDULE (MPS)

MPS is a series of time phased quantities for each item that a company produces, indicating how many are to be produced and when. MPS is initially developed from firm customer orders or from forecasts of demand before MRP system begins to operate. The MRP system whatever the master schedule demands and translates MPS end items into specific component requirements. Many systems make a simulated trial run to determine whether the proposed master can be satisfied.

INVENTORY STATUS FILE

Every inventory item being planned must have an inventory status file which gives complete and up to date information on the on-hand quantities, gross requirements, scheduled receipts and planned order releases for an item. It also includes planning information such as lot sizes, lead times, safety stock levels and scrap allowances.

3. BILL OF MATERIALS (BOM)

BOM identifies how each end product is manufactured, specifying all subcomponents items, their sequence of build-up, their quantity in each finished unit and the work centres performing the build-up sequence. This information is obtained from product design documents, workflow analysis and other standard manufacturing information.

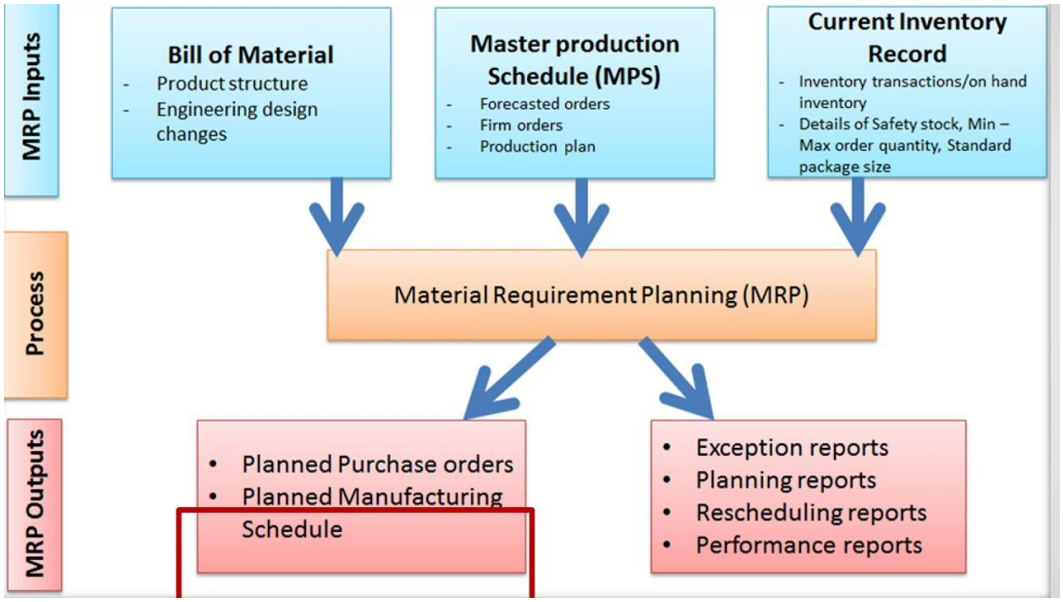
OUTPUTS OF MRP

The outputs of MRP systems dynamically provide the schedule of materials for the future – amount of each material required in each time period to support the MPS. Two primary outputs result:

1. Planned order schedule – a plan of the quantity of each material to be ordered in each time period. This schedule is used by purchasing to place orders with suppliers and by production to order parts, subassemblies, or assemblies from upstream

production departments. The planned orders become guide for future production at suppliers and for in-house production schedules.

2. Change in planned orders – modification of previous planned orders. Quantities of orders can be changed, order can be cancelled, or the orders can be delayed or advanced to different time periods through the updating process.



MRP SYSTEM

MRP LOGIC

Following are some of the terms frequently used on MRP planning forms.

1. **Gross requirements:** Projected needs for raw materials, components, subassemblies, or finished goods by the end of the period shown. Gross requirements come from the master schedule (for end items) or from the combined needs of other items.

2. **Scheduled receipts:** Materials *already on order* from a vendor or in-house shop due to be received *at the beginning* of the period. MRP form shows quantity and projected time of receipt. (***Note:*** Some MRP forms include planned receipts here too.)

3. **On hand/available:** The quantity of an item expected to be available *at the end* of the time period in which it is shown. This includes amount available from previous period plus planned-order receipts and scheduled receipts less gross requirements.

4. **Net requirements***:* Net amount needed in the period. This equals the gross requirements less any projected inventory available from the previous period along with any scheduled receipts.

5. **Planned-order receipt***:* Materials that *will be ordered* from a vendor or in-house shop to be received at the beginning of the period. Otherwise, similar to a scheduled receipt.

6. **Planned-order release:** The planned amount to be ordered in the time period adjusted by the lead-time offset so that materials will be received on schedule. Once the orders are actually released, the planned-order releases are deleted from the form and the planned-order receipts they generated are changed to scheduled receipts.

The MRP Processing Logic/Computer Program

The MRP computer program operates this way:

1. First, with the MPS it begins to determine the number of end items needed in each time period (typically a week). Time periods are sometimes called ***buckets*** in MRP terminology.
2. Next, the numbers of service parts not included in the MPS but deduced from customer orders are included as end items.
3. Next, the MPS and service parts are exploded into gross requirements for all materials by time period into the future by consulting the bill of materials file.
4. Next, the gross materials requirements are modified by the amount of materials on hand and on order for each period by consulting the inventory status file. The net requirements of each material for each bucket are computed as follows:

   Net requirements = Gross Requirements – [Inventory Stock – Safety Stock – Inventory allocated to other uses]

   If the net requirements are greater than zero, orders for the material must be placed. This procedure results in inventory transactions data (orders released, changes in orders, and so on), which are used to update the inventory status file, the primary output reports and secondary output report

Advantages of MRP:

- Materials and components are available when needed
- Minimized inventory levels and associated costs
- Reduced customer lead times
- Increased manufacturing efficiency
- Increased labour productivity

Limitations of MRP:

- Expensive to implement
- Lack of flexibility in the production schedule
- Tendency to hold more inventory than needed
- Less capable than an overall ERP system

ENTERPRISE RESOURCE PLANNING (ERP)

Enterprise resource planning (ERP) is an extension of MRP systems. While MRP is a planning and control system for the resources in a company, ERP is a solution for the enterprise as a whole and an ERP system includes advanced functionality in the areas of financial, customer relationships, and sales order management.

MRP can be a stand-alone application or a piece of an ERP, a single solution that addresses all business needs, not just the scheduling of resources. It decreases any information redundancies and adds elements, like user-level security.

Benefits of an ERP system include increased efficiency, integrated information, customized reports, and higher-quality customer service.

MANUFACTURING RESOURCE PLANNING (MRP II)

MRP II is a computer-based system that can create detailed production schedules using real-time data to coordinate the arrival of component materials with machine and labour availability.

- Manufacturing Resource Planning (MRP II) is an integrated information system used by businesses.
- MRP II is an extension of materials requirement planning (MRP).
- Both MRP and MRP II are seen as predecessors to Enterprise resource planning (ERP).

Overview of MRP II

MRP II is an extension of the original materials requirements planning (MRP I) system. Materials requirements planning (MRP) is one of the first software-based integrated information systems designed to improve productivity for businesses.

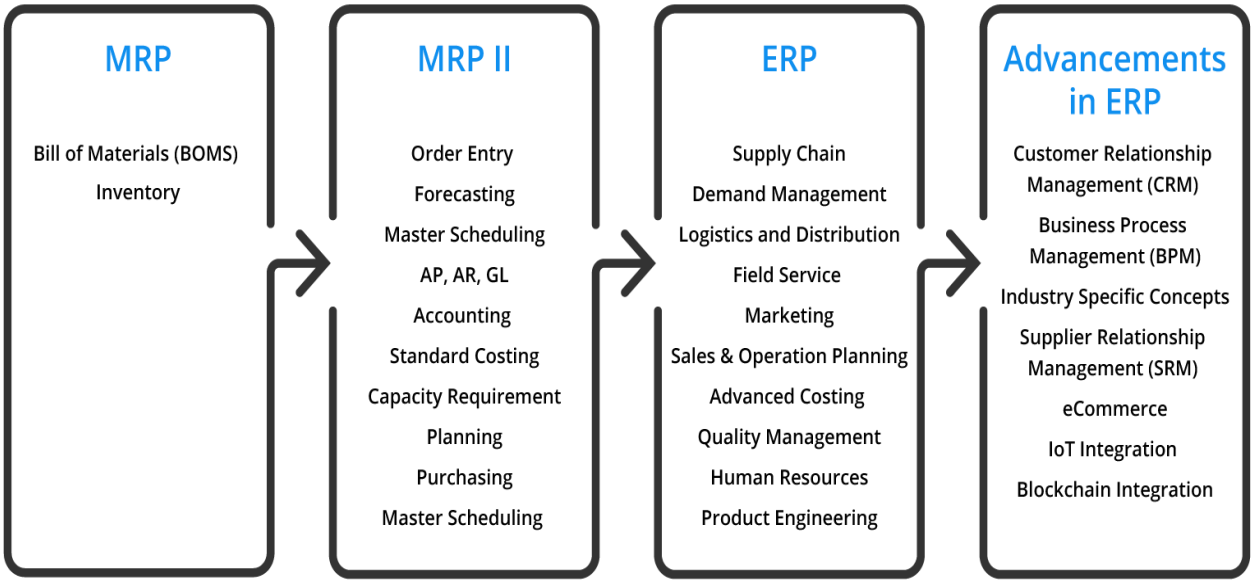
A materials requirements planning information system is a sales forecast-based system used to schedule raw material deliveries and quantities, given assumptions of machine and labour units required to fulfil a sales forecast. At later manufacturers realized they needed software that could also tie into their accounting systems and forecast inventory requirements. MRP II was provided as a solution, which included this functionality in addition to all the capabilities offered by MRP I.

MRP I included the following three major functionalities:

- master production scheduling
- bill of materials
- inventory tracking

MRP II includes those three, plus the following:

- machine capacity scheduling
- demand forecasting
- quality assurance
- general accounting



JUST-IN-TIME (JIT) MANUFACTURING

Just-in-time (JIT) manufacturing, also known as just-in-time production or the Toyota Production System (TPS), was first developed and perfected within the Toyota manufacturing plants.

The just-in-time (JIT) inventory system is a management strategy that aligns raw-material orders from suppliers directly with production schedules. Companies employ this inventory strategy to increase efficiency and decrease waste by receiving goods only as they need them for the production process, which reduces inventory costs. This method requires producers to forecast demand accurately.

The just-in-time (JIT) inventory system minimizes inventory and increases efficiency. JIT production systems cut inventory costs because manufacturers receive materials and parts as needed for production and do not have to pay storage costs. Manufacturers are also not left with unwanted inventory if an order is cancelled or not fulfilled.

One example of a JIT inventory system is a car manufacturer that operates with low inventory levels but heavily relies on its supply chain to deliver the parts it requires to build cars on an as-needed basis. Consequently, the manufacturer orders the parts required to assemble the vehicles only after an order is received.

Advantages of JIT:

1. JIT aims at keeping the stock holding to bare minimum leading to much lower inventory cost and much lower storage and warehouse cost.
2. Minimum inventory at all stages of the supply chain means lesser expiry and lesser wastage for the organization
3. Lower inventory means lower investment for the same level of production. This reduces working capital investment to a greater extent. This in turn leads to better ROI and profitability for the organization
4. JIT manufacturing aims at producing items based on demand. Hence all items produced will be sold. So, no obsolete items in the finished goods section. It helps the organization to adapt well to any changes in product specification from the market without the fear of having any kind of waste and obsolete stocks.
5. To achieve JIT production, there is a need for very close communication between all the parties involved in the entire supply chain. This communication technique is often referred as “Kanban”.
6. In JIT, immense focus is on quality of the final product and companies work to achieve “first time right” for all goods.

Disadvantages of JIT:

1. JIT production can be very sensitive to any kind of error. Since bare minimum inventory levels are maintained, there is no room for any kind of error.
2. JIT production will not be able to adapt well to sudden increase in volume of demand from the market, since the inventory levels are maintained at much lower level.
3. Just-in-time manufacturing is a philosophy which is achieved only when all the parties involved in the whole supply chain will work in great tandem and coordination. JIT may fail sometime if any of the suppliers will fail to fulfil their obligations and respond to the requirements in a timely manner.